higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

ELECTRONICS
NQF LEVEL 2

27 FEBRUARY 2019

This marking guideline consists of 7 pages.

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SECTION A

QUESTION 1

1.1	1.1.1	C		
	1.1.2	C		
	1.1.3	A		
	1.1.4	C		
	1.1.5	B		
			(5 × 1)	(5)
1.2	1.2.1	False		
	1.2.2	True		
	1.2.3	False		
	1.2.4	False		
	1.2.5	True		
			(5 × 1)	(5)
1.3	1.3.1	Solar energy		
	1.3.2	N-type		
	1.3.3	Error light		
	1.3.4	Backplane		
	1.3.5	WAN or Wide Area Network		
			(5 × 1)	(5)
				[15]

TOTAL SECTION A: 15

SECTION B

QUESTION 2

2.1 2.1.1 The wind turns blades, which spin a shaft,✓ which connects to a generator that makes electricity.✓

2.1.2 Electricity is produced from splitting uranium atoms.✓ The heat generated from the split is used to turn turbines which produce electricity.✓

(2 × 2) (4)

2.2 Current is directly proportional to voltage✓ and inversely proportional to resistance in a close circuit.✓ (2)

2.3
$$H = \frac{NI}{L}$$
$$I = \frac{HL}{N} \checkmark$$
$$= \frac{50 \times 12}{800} \checkmark$$
$$= 0{,}75\ A \checkmark$$
(3)

2.4 2.4.1
$$R_T = R_1 + R_2$$
$$= 10 + 300 \checkmark$$
$$= 310\ \Omega \checkmark$$

2.4.2
$$V_T = I_{total} \times R_T$$
$$= 40 \times 10^{-3} \times 310 \checkmark$$
$$= 12{,}4\ V \checkmark$$

2.4.3
$$P_{R_2} = \frac{V_2^2}{R_2}$$
$$= \frac{(I_T \times R_2)^2}{R_2}$$
$$= \frac{(40 \times 10^{-3} \times 300)^2}{300} \checkmark$$
$$= 0{,}48\ W \text{ or } 480\ mW \checkmark$$

OR

$P_{R2} = I^2R$
$= (40 \times 10^{-3})^2 \times 300$✓
$= 480$ mW ✓

(3 × 2) (6)

2.5
- Orange
- Orange
- Red
- Silver

(4)

2.6 2.6.1 Light-dependent resistor (LDR)
2.6.2 Inductor
2.6.3 Photodiode

(3 × 1) (3)

2.7 2.7.1

$$V_T = V_R + V_D$$
$$V_R = V_T - V_D \checkmark$$
$$= 9 - 0{,}7 \checkmark$$
$$= 8{,}3\ V \checkmark$$

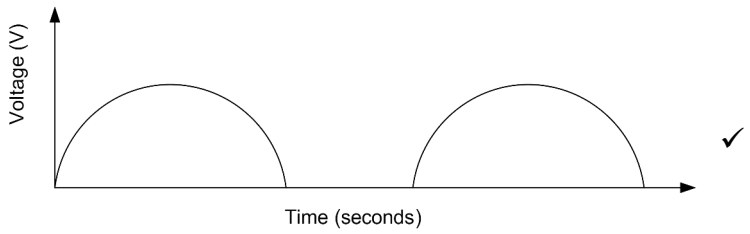
(3)

2.7.2

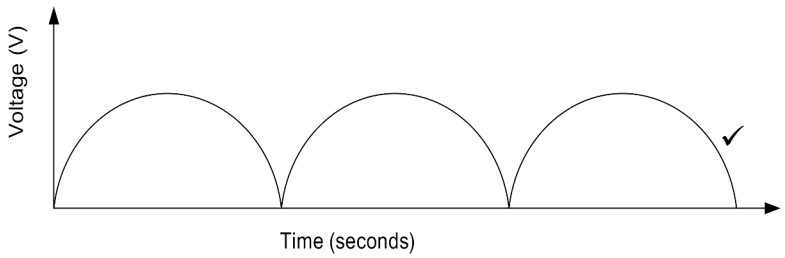
$$I_T = \frac{8{,}3}{470} \checkmark$$
$$= 0{,}01769\ A$$
$$= 17{,}659\ mA \checkmark$$

(2)

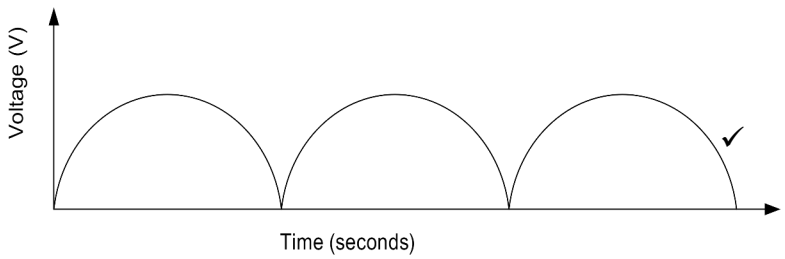
2.8 2.8.1



2.8.2



2.8.3



(3 × 1) (3)
[30]

QUESTION 3

3.1
- Prevents people and animals from being electrocuted due to current leakage
- Enables satisfactory protection under fault conditions by providing low impedance
- Ensures that system voltage remains within reasonable limits
- Ensures that insulation-breakdown voltages are not exceeded
- Complies with regulations and codes of practice (5)

3.2
- Always select the highest scale if unsure of the expected value.✓
- Observe polarity when using an analogue meter.✓ √
- Connect the voltmeter in parallel with the load when measuring voltage✓ and in series with the load when measuring current.✓
- Always consider the loading effect of the voltmeter.✓ (5)

3.3 3.3.1 Tinning is the coating of leads or wires of components with a light coat of solder. (1)

3.3.2
- Clean all dirt or oxidation from all connections.
- Allow solder to melt and spread over the surface of the workplace.
- Heat up the surface with a soldering iron and touch the solder on the surface.
- Use glass cleaner to get rid of any excess solder if reusing resin-core solder. (4)

3.4
- List the tools and equipment needed.
- List the materials needed.
- List the safety equipment needed.
- List what to do to prepare the workshop area for the job.
- List the safety precautions needed. (5) √

[20]

QUESTION 4

4.1 4.1.1 $= 2^3 \times 1 + 2^2 \times 1 + 2^1 \times 0 + 2^0 \times 1$ ✓
$= 13_{10}$ ✓

4.1.2 $= 2^4 \times 1 + 2^3 \times 1 + 2^2 \times 0 + 2^1 \times 0 + 2^0 \times 1$ ✓
$= 25_{10}$ ✓

(2 × 2) (4)

4.2 4.2.1

$$\begin{array}{r} {\scriptstyle 1\quad\quad 1\quad\quad} \\ 1\,1\,0\,1 \\ +\;1\,1\,0\,0\,1 \\ \hline 1\,0\,0\,1\,1\,0 \end{array}$$ ✓✓

4.2.2

$$\begin{array}{r} {\scriptstyle 1\ \ 1} \\ 1\,1\,0\,0\,1 \\ -\ 1\,1\,0\,1 \quad \checkmark \\ \hline 1\,1\,0\,0 \quad \checkmark \end{array}$$

(2 × 2) (4)

4.3 4.3.1 OR gate (1)

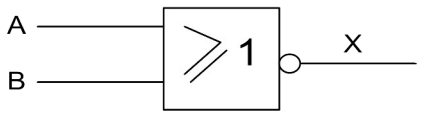
4.3.2

Switch A	Switch A	Output X	
0	0	0	✓
0	1	1	✓
1	0	1	✓
1	1	1	✓

(4)

4.3.3 $X = A + B$ (1)

4.3.4



(1)
[15]

QUESTION 5

5.1
- Install relevant software to create or change the program.✓
- The program must come with a set of standard instructions✓ to use the software so that the processor understands what the programmer wants.✓
- Run the software either off-line or online.✓
- Use the connector cable to connect the computer and the PLC to transfer data from the programmer to the processor.✓ (5)

5.2 Multiplexing is the process that combines two or lower bandwidth transmissions into one higher bandwidth.✓✓ It splits the total available bandwidth into narrower bands✓ and divides a channel to several transmitting sources one at the time.✓ Therefore the bandwidth or amount of information that can be transmitted at a certain time is increased.✓ (5)

5.3

AMPLITUDE MODULATION (AM)	FREQUENCY MODULATION (FM)
•It is more prone to interference. •It has poor sound quality. •It can be transmitted over long distances. •It is cheaper. •The transmitter and receiver are simple. •Carrier wave is modulated in the amplitude, i.e. the frequency and the phase remain the same during modulation.	• It is less prone to interference. • It has better sound quality. • It cannot be transmitted over long distances. • It is expensive. • The transmitter and receiver are complex. • Carrier wave is modulated in the frequency, i.e. the amplitude and the phase remain the same during modulation.

(Any 3 + any 3) (6)

5.4 •A local area network (LAN)✓ is a computer network that covers a local area.✓
•A wide area network (WAN) is a computer network that covers a wide area.✓
• The internet✓ is a computer network that connects computers and other devices to the World Wide Web.✓ (Any 2 × 2) (4)

[20]

TOTAL SECTION B: 85
GRAND TOTAL: 100

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